WORKSHEET

Chapter 6: Finding definite integrals

1 Calculate each of the following definite integrals.

a $\int_1^7 x^2\,dx$

b $\int_1^7 \frac{1}{x^2}dx$

c $\int_{-3}^3 (x^3 - x)dx$

d $\int_{-2}^2 (3x^2 - 5)dx$

e $\int_{-10}^{10} (x - 3x^2 + 6x^5)dx$

f $\int_0^{36} \frac{1}{2}\sqrt{x}\,dx$

g $\int_0^{10} e^{-x}\,dx$

h $\int_0^2 e^{-3x}\,dx$

2 CAS Evaluate parts **b**, **d** and **f** of question **1** using a CAS calculator.

© Cengage Learning Australia Pty Ltd 2014

3 Evaluate each of the following, giving the answer in exact form.

a $\int_0^1 e^{5x}\,dx$

b $\int_0^2 -e^{-x}\,dx$

c $\int_1^4 2e^{3x+4}\,dx$

d $\int_2^3 (3x^2 - e^{2x})dx$

e $\int_0^2 (e^{2x} + 1)dx$

f $\int_1^2 (e^x - x)dx$

g $\int_0^3 (e^{2x} - e^{-x})dx$

4 Evaluate correct to 2 decimal places.

a $\int_1^3 e^{-x}\,dx$

b $\int_0^2 2e^{3x}\,dx$

c $\int_5^6 (e^{x+5} + 2x - 3)dx$

© Cengage Learning Australia Pty Ltd 2014

d $\int_0^1 (e^{3x+4} - x)dx$

e $\int_1^2 (e^{4x} + e^{2x})dx$

5 CAS Evaluate parts **a**, **c** and **e** of question **4** using a CAS calculator.

6 Evaluate each of the following.

a $\int_0^2 (3 - 2x)^5 \, dx$

b $\int_0^1 \frac{(3x - 1)^2}{6} \, dx$

c $\int_4^5 (5 - x)^6 \, dx$

d $\int_3^6 \sqrt{x - 2} \, dx$

e $\int_0^1 \frac{dx}{(3x - 2)^4}$

f $\int_0^2 \frac{5}{(2x + 1)^3} dx$

g $\int_1^4 \frac{2}{\sqrt{(5x - 4)^3}} dx$

© Cengage Learning Australia Pty Ltd 2014

7 CAS Evaluate parts **b**, **d** and **f** of question **6** using a CAS calculator.

8 Evaluate each of the following, giving exact answers where appropriate.

a $\int_0^{\frac{\pi}{2}} \cos(x)\,dx$

b $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sec^2(x)\,dx$

c $\int_{\frac{\pi}{2}}^{\pi} \sin\left(\frac{x}{2}\right)dx$

d $\int_0^{\frac{\pi}{2}} \cos(3x)\,dx$

e $\int_0^{\frac{1}{2}} \sin(\pi x)\,dx$

f $\int_0^{\frac{\pi}{8}} \sec^2(2x)\,dx$

g $\int_0^{\frac{\pi}{12}} 3\cos(2x)\,dx$

h $\int_0^{\frac{\pi}{10}} -\sin(5x)\,dx$

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** 114

b $\frac{6}{7}$

c 0

d -4

e -2000

f 72

g $1 - e^{-10}$

h $\frac{1}{3} - \frac{1}{3}e^{-6} \approx 0.3325$

3 **a** $\frac{1}{5}(e^5 - 1)$

b $\frac{1}{e^2} - 1$

c $\frac{2}{3}e^7(e^9 - 1)$

d $19 - \frac{1}{2}e^4(e^2 - 1)$

e $\frac{1}{2}e^4 + 1\frac{1}{2}$

f $e^2 - e - 1\frac{1}{2}$

g $\frac{1}{2}e^6 + e^{-3} - 1\frac{1}{2}$

4 **a** 0.32

b 269.29

c 37 855.68

d 348.85

e 755.19

6 **a** $60\frac{2}{3}$

b $\frac{1}{6}$

c $\frac{1}{7}$

d $4\frac{2}{3}$

e $-\frac{1}{8}$

f $1\frac{1}{5}$

g $\frac{3}{5}$

8 **a** 1

b $\sqrt{3} - \frac{1}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$

c $\frac{2}{\sqrt{2}} = \sqrt{2}$

d $-\frac{1}{3}$

e $\frac{1}{\pi}$

f $\frac{1}{2}$

g $\frac{3}{4}$

h $-\frac{1}{5}$

© Cengage Learning Australia Pty Ltd 2014